

State of the art

Title: Design and Evaluation of an LLM-based Retrieval-Augmented-Generation Prototype for Security Analyst Support in Cyber Threat Intelligence

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1. Introduction

Cyber Threat Intelligence (CTI) analysis remains highly manual because threat data is heterogeneous, complex, and often unstructured (Alhuzali, 2025). CTI only provides operational value when it is delivered in a timely, relevant, and accurate manner, which is difficult to achieve under time pressure (Palacín, 2021). While standalone LLMs can support analysis, they are prone to hallucinations, may rely on outdated knowledge in fast-evolving vulnerability landscapes, and can misinterpret technical content (Alhuzali, 2025). A further challenge is traceability: purely generative outputs often lack transparent provenance, complicating expert verification (Rajapaksha et al., 2024). Retrieval-Augmented Generation (RAG) is therefore studied as a mitigation, as it links generation to external knowledge bases and can provide grounded answers with verifiable citations (Agnihotram and Sarkar, 2025; Rajapaksha et al., 2024).

1.1 Topic and objectives

This State-of-the-Art focuses on LLM-based Retrieval-Augmented Generation (RAG) systems designed to support security analysts in CTI tasks, particularly in contexts such as cyber-attack investigation and attribution where verification is essential (Rajapaksha et al., 2024). The reviewed work indicates that current research frequently targets specific intelligence tasks such as the interpretation of tactics, techniques and procedures (TTPs) and the extraction of indicators of compromise (IOCs) from reports (Fayyazi et al., 2024). At the same time, proposed systems differ in how they operationalize grounding—ranging from context-supported generation to explicit citation-based outputs intended to reduce hallucinations and enable efficient verification (Agnihotram and Sarkar, 2025; Rajapaksha et al., 2024).

1.2 Literature review methodology

The literature for this State of the Art was collected through a structured search and screening process. The search focused on publications from 2019 to 2025 and targeted major academic databases and preprint repositories, including ACM Digital Library, IEEE Xplore, ScienceDirect, Web of Science, SpringerLink, Wiley Online Library, and arXiv. Search strings combined LLM-related terms (e.g., *large language model*, *GPT*, *retrieval-augmented generation*, *RAG*) with CTI- and SOC-relevant keywords (e.g., *threat intelligence*, *incident response*, *vulnerability*, *threat hunting*).

Studies were included when they provided sufficient methodological detail and contributed directly to at least one of the following aspects: (i) LLM/RAG-based support for CTI or SOC analyst tasks, (ii) architectural design choices for retrieval and evidence-grounded generation, or (iii) evaluation approaches for RAG systems in security-relevant settings. Publications were excluded if they were too short to provide substantive technical content, focused exclusively on traditional machine learning without LLMs, or addressed only the security of LLMs without relevance to CTI analyst support. Screening was performed in multiple stages (title/abstract followed by full-text review), and references of selected papers were inspected to identify additional relevant work.

Database / Source	Search terms	Date	Initial results (approx.)	Selected for review
Google Scholar	"RAG LLM"	12.12.2025	>5,000	8
Google Scholar	"augmented LLM"	12.12.2025	>8,000	6
Google Scholar	"LLM with external knowledge base"	12.12.2025	>6,000	5
ACM Digital Library	"retrieval-augmented generation" AND "threat intelligence"	15.12.2025	~120	6
IEEE Xplore	"large language model" AND "security operations"	15.12.2025	~95	5
ScienceDirect	"RAG" AND "cyber security"	15.12.2025	~80	4
Web of Science	"LLM" AND "incident response"	17.12.2025	~60	3
SpringerLink	"generative AI" AND "cyber threat intelligence"	21.12.2025	~70	4
Wiley Online Library	"large language models" AND "SOC analysis"	5.1.2026	~45	2
arXiv (preprints)	"retrieval augmented generation" AND "cybersecurity"	7.1.2026	~150	6

Table 1: Overview of the structured literature search process, including databases, representative search terms, search period, and screening outcomes

1.3 Delimitations of this literature overview

The focus for this review lies on approaches that explicitly combine retrieval mechanisms with generative models to support analyst-facing tasks such as investigation, attribution, triage, and incident response decision-making. Several related research areas are out of scope. In particular, this review does not cover malware reverse engineering, IDS/IPS detection modelling, signature-based intrusion detection, or purely predictive machine learning approaches that do not involve retrieval-augmented generation. Likewise, work focusing exclusively on securing LLMs (e.g., prompt injection defense without CTI application) is excluded unless it directly affects analyst-facing RAG pipelines.

From a methodological perspective, the review concentrates on architectural design choices and evaluation approaches rather than large-scale user studies. While human factors such as trust and verification effort are discussed conceptually, extensive empirical user research is not systematically reviewed. Finally, due to language and data availability constraints, the literature primarily reflects English-language sources and publicly accessible CTI corpora, which may limit the generalizability of some findings.

2. Theoretical background

2.1 Cyber threat intelligence and threat hunting fundamentals

Cyber Threat Intelligence (CTI) can be described as a systematic process of collecting and analyzing actionable insights about current or potential threats in order to support proactive security decision-making (Alhuzali, 2025). In practice, CTI is communicated through artefacts such as indicators of compromise (IOCs), malware signatures, and structured behavioral patterns (TTPs). To make such artefacts interoperable and reusable across teams and tools, they are frequently organized using frameworks and standards such as MITRE ATT&CK (Alhuzali, 2025; Rajapaksha *et al.*, 2024).

Threat hunting complements CTI by taking a proactive stance: it aims to identify unknown or unresolved threats inside an environment before they cause harm (Sun *et al.*, 2023). This requires analysts to synthesize and contextualize isolated indicators, connecting technical traces to broader hypotheses about attacker behavior, trends, and potential motivations (Rastogi and Alam, 2023). As a result, analyst work is shaped by two recurring constraints: the need for timeliness and the need for verifiability. Timeliness matters because static model knowledge can fail to capture newly emerging vulnerabilities—particularly zero-days—without mechanisms to incorporate fresh intelligence (Paul *et al.*, 2025).

Finally, CTI analyst support systems must enable efficient and defensible workflows. Traceability through source-based evidence is essential to mitigate hallucinations and reduce manual verification effort, especially in investigation and attribution scenarios (Rajapaksha *et al.*, 2024). At the same time, heterogeneous CTI sources vary in credibility; therefore, systems must account for source reliability to avoid propagating misinformation, including low-quality or AI-generated content (Hwang *et al.*, 2025). Against this background, RAG-based approaches are motivated not only by the promise of faster triage of unstructured reports, but also by the ability to shorten response times while maintaining verifiable, evidence-linked outputs (Tseng *et al.*, 2024).

2.2 LLM and retrieval-augmented generation fundamentals

Large Language Models (LLMs) are transformer-based models that can support CTI analysts through natural language interaction and by summarizing or extracting actionable information from large volumes of unstructured data (Divakaran and Peddinti, 2024). However, their application in security contexts is constrained by recurring risks, notably hallucinations, limited recency due to static parametric knowledge, and missing provenance for generated claims (Alhuzali, 2025).

Retrieval-Augmented Generation (RAG) mitigates these limitations by combining the LLM with an external, non-parametric knowledge store via a pipeline that typically includes ingestion, chunking, indexing, retrieval, reranking, context assembly, and generation (Gao *et al.*, 2024). Retrieval may be sparse (e.g., BM25), dense (vector similarity), or hybrid to balance precision and recall, with reranking used to prioritize the most relevant evidence for generation (Agnihotram and Sarkar, 2025; Gao *et al.*, 2024). For analyst-facing CTI support, grounding and faithfulness are central: outputs should remain consistent with retrieved context and be supported by evidence, ideally with explicit citations to reduce verification effort (Es *et al.*, 2025; Rajapaksha *et al.*, 2024). At the same time, common failure points include inaccurate retrieval, context overload, and misattribution of claims to sources (Yu *et al.*, 2025).

2.3 Evaluation foundations for RAG systems

Evaluating RAG systems is inherently multi-component because retrieval and generation contribute different failure modes and therefore need to be assessed separately (Es *et al.*, 2025). On the retrieval side, common metrics include Precision@k to quantify how much of the retrieved set is relevant, Recall@k to capture completeness, and rank-aware measures such as MRR or nDCG to reflect whether the most relevant evidence appears early in the ranking (Agnihotram and Sarkar, 2025). On the generation side, evaluation often distinguishes between *faithfulness/groundedness*—the extent to which answers remain supported by retrieved evidence—and *answer correctness*, which compares outputs to reference answers where such references exist (Yu *et al.*, 2025).

For analyst-facing use cases, automated evaluation can improve scalability, but “LLM-as-a-judge” approaches may introduce biases and may not align with expert judgement in specialized domains (Akkiraju *et al.*, 2024). This limitation is amplified in CTI, where robust ground truth is frequently hard to obtain for complex tasks such as TTP mapping, constraining the feasibility of reference-based evaluation (Fayyazi *et al.*, 2024). As a result, verifiability becomes a key practical criterion: systems should provide source-based citations that support traceability and reduce manual verification effort (Rajapaksha *et al.*, 2024; Mangold and Hoffmann, 2025).

3. Current state of research

3.1 Challenges in CTI analyst workflows

CTI analysts operate in an environment characterized by information overload and heterogeneous inputs. The volume and non-uniform, often informal structure of CTI feeds can exceed analysts’ capacity to process them consistently, and fully automated processing remains difficult in practice (Rastogi and Alam, 2023). As a result, substantial time is spent reading and interpreting natural-language CTI reports, which delays response activities and reduces the time available for higher-value investigative work (Tseng *et al.*, 2024). This pressure is amplified by operational constraints in SOC settings, including limited resources and gaps in process integration and automation (Ismail *et al.*, 2025).

A second set of challenges concerns reliability and context. Static approaches struggle to keep pace with rapidly evolving threats and the continuous emergence of new vulnerabilities, making timeliness a persistent requirement for analyst support (Paul *et al.*, 2025). At the same time, analysts need verifiability: without transparent provenance, purely generative outputs are difficult to validate, which is problematic for high-stakes tasks such as investigation and attribution (Rajapaksha *et al.*, 2024). Source credibility adds further complexity, as systems may ingest and propagate misinformation from unreliable sources, including low-quality or AI-generated content (Hwang *et al.*, 2025). Finally, conventional tooling often provides outputs that are difficult to interpret—such as static IDS log strings—without interactive clarification or deeper contextualization (Huang *et al.*, 2024), while external CTI can remain too abstract if it does not reflect local system state and organizational policies (Rastogi and Alam, 2023).

3.2 LLM use in cybersecurity and threat intelligence

3.2.1 Evidence from surveys and empirical studies

Survey and review evidence suggests that LLM applications in cybersecurity concentrate on a set of recurring task categories, most prominently vulnerability-related analysis, malware analysis, and

support for intrusion response activities (Xu *et al.*, 2025). Across these areas, LLMs are commonly framed as productivity enablers: they can automate repetitive subtasks and lower the entry barrier for analysts by making security knowledge more accessible through natural-language interaction (Divakaran and Peddinti, 2024).

At the same time, the literature highlights failure modes that are particularly consequential in security settings. Reported issues include factual hallucinations, recency limitations due to static training data, and limited traceability of the reasoning behind model outputs (Gao *et al.*, 2024). Empirical evidence remains constrained: end-to-end evaluations embedded in realistic SOC workflows are still rare, with many studies relying on isolated lab environments or synthetic datasets, which limits external validity for operational CTI use (Tseng *et al.*, 2024).

3.2.1 CTI task support patterns

Across CTI-focused applications, the literature shows several recurring patterns in how LLM-based systems are used to support analysts. A first pattern is structured summarization of unstructured threat data: LLMs are applied to extract key variables from forum and report-like content, turning narrative text into analyst-relevant fields that can be reviewed and compared (Clairoux-Trepanier *et al.*, 2024). A second pattern is IOC and TTP extraction for downstream operationalization. Here, model outputs are not limited to plain text; instead, extracted indicators such as registry keys or IP addresses can be transformed into structured detection artefacts (e.g., regex rules) that are usable in SIEM contexts (Tseng *et al.*, 2024).

A third pattern targets investigation and attribution support through evidence-linked question answering. In such setups, systems aim to reduce ambiguity by providing verifiable citations to the underlying sources used to justify attribution-relevant statements (Rajapaksha *et al.*, 2024). Related work extends analyst support into incident response guidance, where LLM outputs are framed as playbook-style actions derived from event logs and aligned with established frameworks such as NIST CSF and MITRE ATT&CK (Ismail *et al.*, 2025). Beyond advisory outputs, some approaches attempt policy-to-action automation by decomposing high-level security policies into executable actions, such as Windows API calls (Saura *et al.*, 2025).

More recent designs introduce agentic workflows, where autonomous LLM agents iteratively refine classifications and produce SOC-ready reports rather than performing a single-pass generation step (Huang *et al.*, 2024).

3.3 RAG for CTI: Design space and architectures

3.3.1 Retrieval strategies: sparse, dense, hybrid and reranking

Retrieval is commonly implemented as sparse keyword search, dense vector retrieval, or a hybrid of both. BM25-style sparse retrieval scales well but can suffer from vocabulary mismatch (Robertson and Zaragoza, 2009). Dense retrieval mitigates this by using embeddings to capture semantic similarity (Agnihotram and Sarkar, 2025). Hybrid retrieval combines both signals and is often motivated in CTI settings where terminology and reporting styles vary across sources (Alhuzali, 2025). Many pipelines add reranking with cross-encoders to prioritize the most relevant evidence for the LLM context (Gao *et al.*, 2024). Increasing top-*k* can raise recall but may also introduce noise, reducing precision and potentially increasing downstream hallucination risk; quality gains therefore come with latency and compute costs, especially in hybrid + reranking setups (Agnihotram and Sarkar, 2025).

3.3.2 Document processing: chunking and metadata for CTI corpora

Chunking determines the granularity of retrieval and is non-trivial for heterogeneous, unstructured CTI reports (Rastogi and Alam, 2023). More structured approaches use semantic or hierarchical chunking to preserve coherence (e.g., topic similarity and entity continuity) and reduce information loss (Wang *et al.*, 2025). Practical pipelines may also apply recursive splitting into larger and smaller chunks to balance broad context with embedding-friendly granularity (Alhuzali, 2025). Metadata enrichment (e.g., CVE IDs, source labels, timestamps) enables filtering and time-aware retrieval, while credibility-related metadata can help reduce exposure to low-quality or misleading sources (Gao *et al.*, 2024; Hwang *et al.*, 2025).

3.3.3 Context assembly and grounded generation

RAG systems assemble top-ranked chunks into the prompt to compensate for gaps in the LLM's parametric knowledge (Gao *et al.*, 2024). In CTI workflows, explicit citations support traceability and verifiable reasoning, particularly for investigation and attribution tasks (Rajapaksha *et al.*, 2024). Some approaches incorporate abstention ("I don't know") to avoid fabricated answers under insufficient evidence (Paul *et al.*, 2025). Common failure modes include context overload ("lost-in-the-middle"), misattribution of claims to sources, and knowledge poisoning when false content enters the underlying corpus, motivating stricter context control and stronger ingestion/verification practices (Gao *et al.*, 2024; Es *et al.*, 2025; Alhuzali, 2025).

3.4 Evaluation approaches for RAG systems

3.4.1 Retrieval evaluation

Retrieval performance in RAG systems is commonly assessed using information-retrieval metrics such as Precision@k and Recall@k (Agnihotram and Sarkar, 2025). Rank-aware measures including MRR and nDCG are used to capture whether the most useful evidence appears early in the ranking (Yu *et al.*, 2025). A recurring pitfall is *top-k noise*: increasing *k* typically raises recall but can reduce precision by adding irrelevant context, which may increase hallucination risk downstream (Agnihotram and Sarkar, 2025). Moreover, traditional IR metrics do not always reflect how useful retrieved passages are for the generation step, creating a metric–utility mismatch in end-to-end RAG evaluation (Yu *et al.*, 2025). To mitigate this, multi-stage pipelines with rerankers are often recommended to prioritize the most relevant evidence before context assembly (Agnihotram and Sarkar, 2025).

3.4.2 Generation evaluation

Generation evaluation in RAG systems focuses less on fluency and more on whether answers are *supported* by retrieved evidence. The literature commonly separates (i) faithfulness/groundedness, i.e., consistency with and attribution to the retrieved context, from (ii) answer quality, such as relevance to the query and correctness against reference answers where available (Yu *et al.*, 2025; Es *et al.*, 2025). This distinction matters because responses can appear plausible while remaining weakly grounded, or they can be grounded yet fail to satisfy the user's intent.

Automated frameworks operationalize these dimensions in different ways. "RAGAS" evaluates a "RAG triad" (context relevance, groundedness, answer relevance) using LLM prompting, enabling largely reference-free scoring when gold answers are scarce (Es *et al.*, 2025). "ARES" uses LLM judges, but

it treats their outputs as potentially noisy and therefore reports confidence intervals to indicate how stable the automated scores are (Saad-Falcon *et al.*, 2024).

For analyst-facing CTI, citation quality and verifiability become central criteria. Yu *et al.* (2025) discuss attributable-to-source evaluation (e.g., AIS-style measures), while Rajapaksha *et al.* (2024) emphasize that explicit citations can reduce manual verification effort in investigation and attribution tasks. However, “LLM-as-a-judge” approaches remain limited by potential bias, insufficient domain expertise, and circularity risks, suggesting that automated metrics should be treated as supportive indicators rather than definitive measures of operational quality (Akkiraju *et al.*, 2024).

3.4.3 Scenario-based evaluation

Scenario-based evaluation complements offline metrics by assessing operational utility within end-to-end workflows. For example, SOC-oriented studies simulate attack and defense scenarios to evaluate how systems support response processes under realistic conditions (Ismail *et al.*, 2025). Typical measures include latency, the accuracy of triage or mitigation recommendations, and alignment with incident response playbooks (Ismail *et al.*, 2025). Some approaches add qualitative expert assessment of report structure, interpretability, and usefulness, for instance via questionnaires (Huang *et al.*, 2024; Mangold and Hoffmann, 2025). Finally, task-specific exam generation has been proposed as a way to construct domain-relevant evaluation questions from CTI corpora and obtain more robust accuracy estimates than generic benchmarks (Guinet *et al.*, 2024).

3.5 Limitations of existing research

A recurring limitation across analyst-support RAG research is the tension between *methodological ambition* and *operational feasibility*. Several approaches rely on multi-stage retrieval and reranking to improve relevance, yet these steps can introduce non-trivial latency, which limits suitability for time-sensitive CTI use cases (Agnihotram and Sarkar, 2025).

A second set of constraints concerns evaluation validity and generalizability. The evidence base remains limited by scarce industrial datasets, which makes it difficult to assess how well results from lab settings transfer to real SOC operations (Xu *et al.*, 2025). Ground truth is similarly hard to establish for CTI tasks such as mapping procedures to MITRE ATT&CK tactics, which constrains robust benchmarking and comparative evaluation (Fayyazi *et al.*, 2024). Where automated evaluation is used, LLM-as-a-judge introduces risks such as circularity and bias amplification (Akkiraju *et al.*, 2024).

Finally, reliability and safety risks are not fully resolved. CTI corpora and external knowledge bases can be incomplete and uneven in coverage, and RAG pipelines remain exposed to knowledge poisoning if manipulated content enters the retrieval store (Alhuzali, 2025). More fundamentally, LLMs may over-trust retrieved context—even when it contains misinformation—highlighting robustness issues that are especially problematic in open-source CTI settings (Chen *et al.*, 2023).

4. Research gaps

4.1 Open issues

A second gap is the limited emphasis on end-to-end SOC workflow evaluation. Existing evaluations often remain centered on controlled benchmarks and isolated technical metrics, while evidence from realistic SOC settings under time pressure is still rare. This constrains what can be concluded about operational usefulness for analysts (Alhuzali, 2025; Mangold and Hoffmann, 2025).

A third gap is the insufficient handling of source reliability and integrity in CTI-RAG. Heterogeneous intelligence feeds differ widely in credibility, yet many approaches do not proactively assess trustworthiness or defend against manipulation, leaving systems vulnerable to misinformation and knowledge poisoning (Hwang *et al.*, 2025; Alhuzali, 2025)

4.2 Implications for this thesis

These gaps imply that an analyst-facing CTI RAG prototype should be designed not only for answer quality, but for verifiability, timeliness, and robustness to unreliable sources, and should be evaluated in a way that better reflects CTI work conditions.

- **Evidence-linked outputs:** The prototype should provide explicit, source-based citations for key claims to support traceability and reduce manual verification effort in investigative tasks (Rajapaksha *et al.*, 2024).
- **Hybrid retrieval as a baseline:** Given CTI terminology and reporting variability, a hybrid approach combining lexical and semantic retrieval is justified to improve the precision–recall balance (Agnihotram and Sarkar, 2025; Alhuzali, 2025).
- **Source integrity and reliability checks:** The pipeline should include mechanisms to estimate and use source reliability signals to reduce the risk of retrieving and propagating misinformation (Hwang *et al.*, 2025; Ammann *et al.*, 2025).
- **Scenario-based evaluation:** Evaluation should move beyond offline metrics and test measures such as task success, latency, and quality of recommendations (Ismail *et al.*, 2025; Huang *et al.*, 2024).
- **Analyst-oriented presentation:** Outputs should be structured to match analyst intent and expertise level, rather than optimized solely for generic quality scores (Mangold and Hoffmann, 2025; Lyu *et al.*, 2024).

5. Conclusion and outlook

5.1 Summary of key findings and outlook

The reviewed literature suggests a converging set of design choices for analyst-facing CTI RAG systems. A recurring best-practice pattern is hybrid retrieval that combines lexical search (e.g., BM25) with semantic vector retrieval to improve the precision–recall balance in CTI settings (Agnihotram and Sarkar, 2025). In parallel, recent work highlights the potential of agentic RAG architectures, which use iterative reasoning and refinement loops to handle more complex analysis tasks than linear pipelines (Gao *et al.*, 2024).

Despite this progress, key limitations remain. A major bottleneck is the lack of publicly available ground-truth datasets for specialized CTI tasks such as mapping attack procedures, which constrains robust evaluation and comparability across approaches (Fayyazi *et al.*, 2024). In addition, source integrity and reliability remain unresolved challenges: RAG pipelines can propagate misinformation when retrieval surfaces low-quality or manipulated content, making credibility assessment a central requirement for operational use (Hwang *et al.*, 2025).

5.2 Critical reflection of the reviewed literature

The reviewed literature demonstrates substantial progress in the application of LLM- and RAG-based systems for CTI analyst support, particularly with respect to retrieval architectures and evidence-grounded generation. However, this progress is accompanied by structural limitations that affect the interpretability and practical relevance of reported results. A recurring issue is the strong focus on technical optimization and proxy metrics, while the broader organizational and operational context of SOC environments is often simplified or abstracted away.

From a methodological perspective, many studies rely on controlled experimental setups and synthetic or narrowly scoped datasets. While such settings enable reproducibility and systematic comparison, they also limit the generalizability of findings to real-world CTI workflows characterized by time pressure, incomplete data, and heterogeneous sources. As a result, improvements in retrieval or generation quality do not necessarily translate into measurable gains in analyst effectiveness or decision quality.

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